

Machine Learning

Lecture 8: Knowledge distillation

Young & Yandex     ШАД



Radoslav Neychev



Outline

01

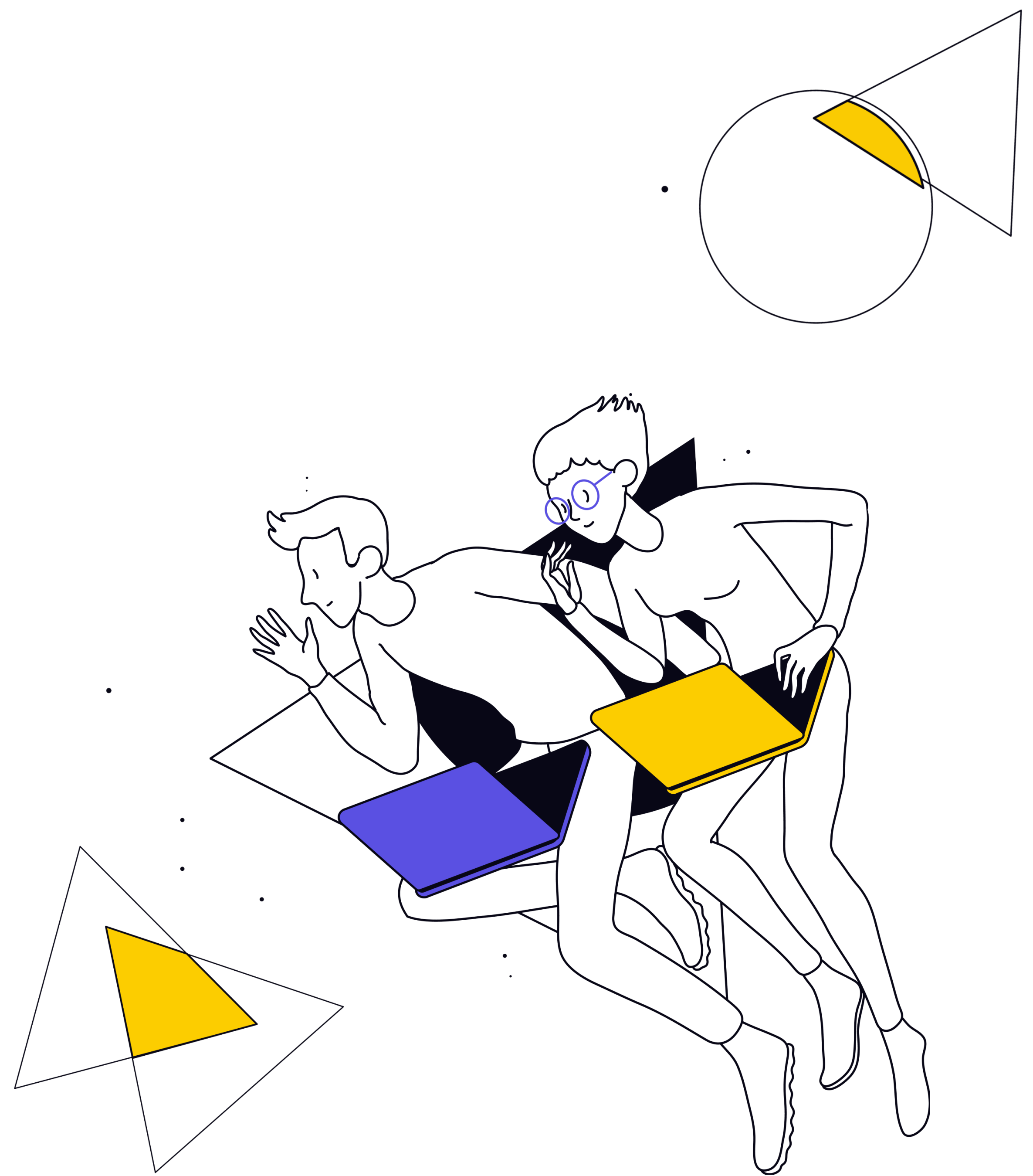
Knowledge distillation

02

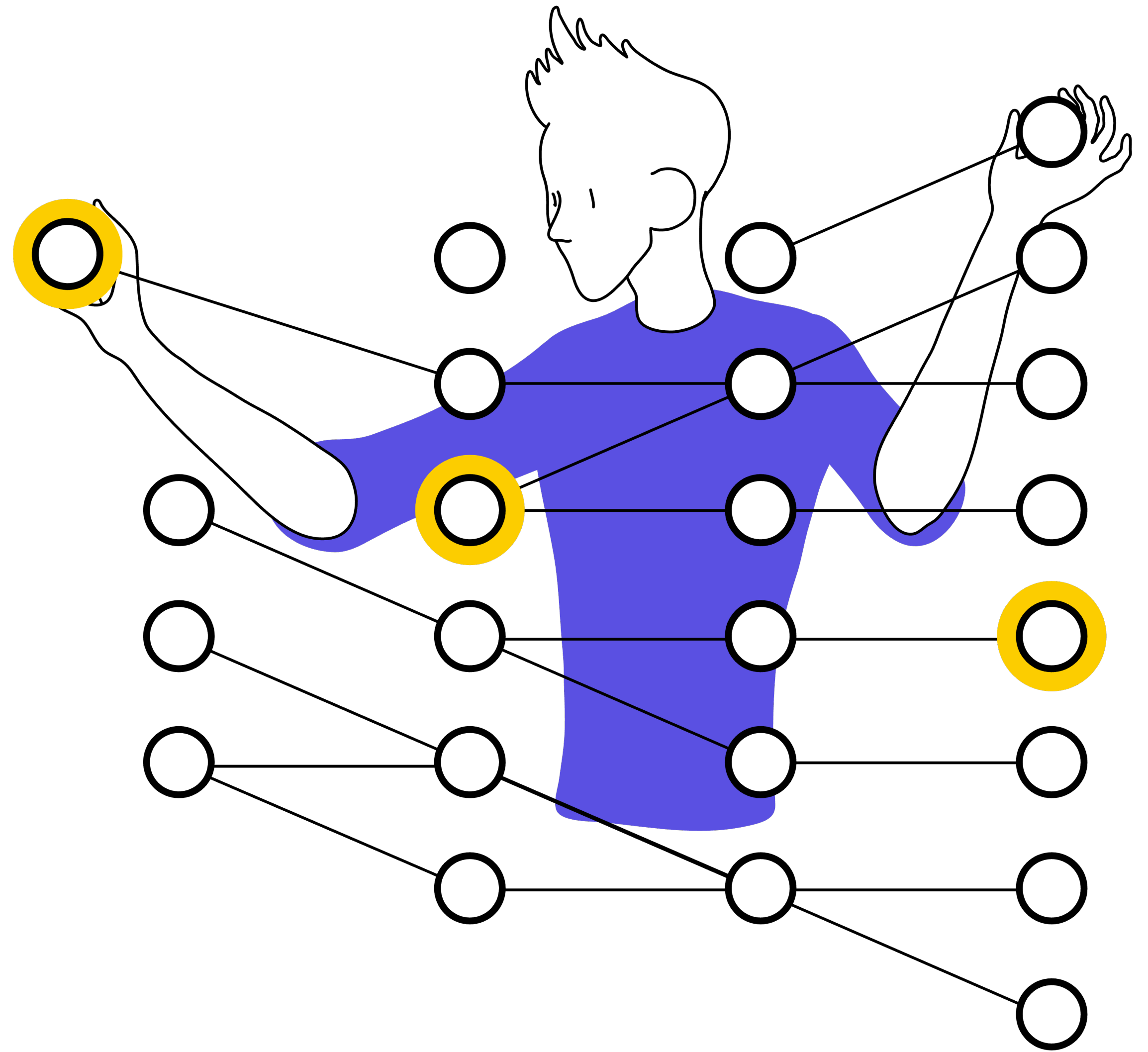
General overview of supervised learning models

03

Outro



Knowledge distillation



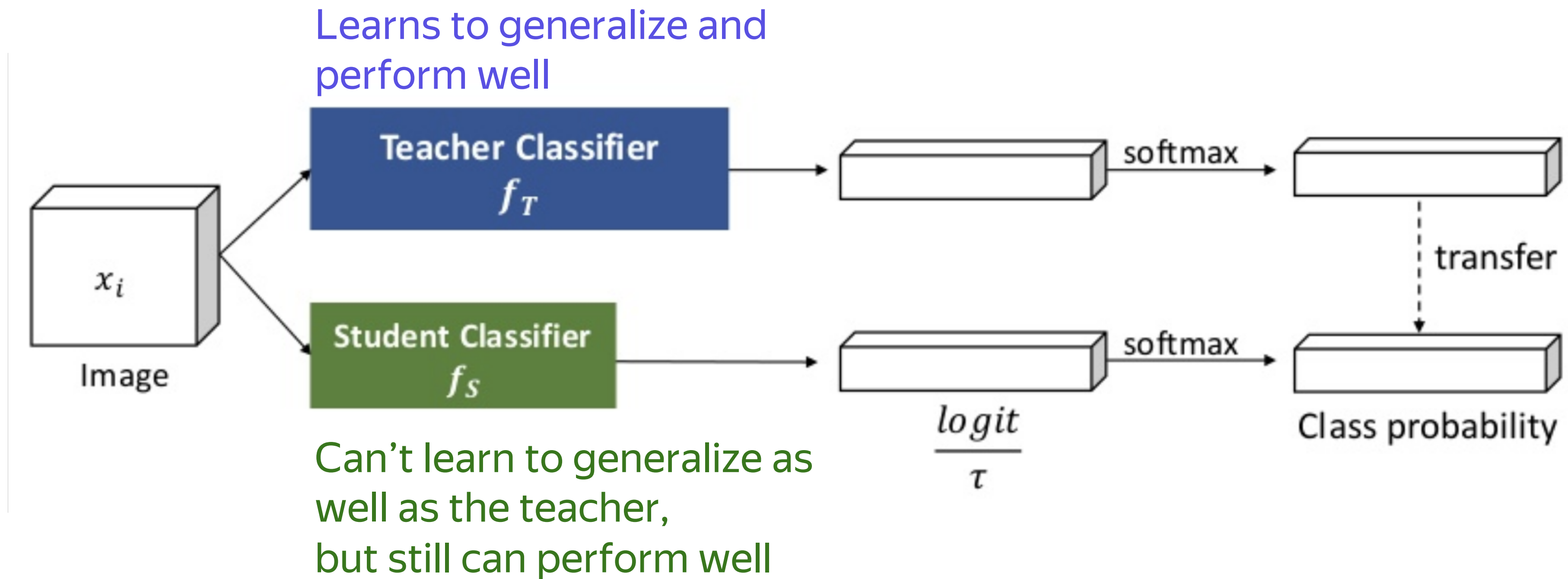
01

Knowledge distillation



Do they have the same “life purpose”
and solve the same problems?

Knowledge distillation



Knowledge distillation

Denote teacher and student models.

Student model has logits z_i and corresponding probabilities q_i , derived with the softmax operation:

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

where T stays for the temperature.

Teacher model has logits v_i and corresponding probabilities p_i .

Knowledge distillation

Let's derive the cross-entropy gradient on **student** logits using the **teacher** predictions as targets:

$$\frac{\partial C}{\partial z_i} = \frac{1}{T} (q_i - p_i) = \frac{1}{T} \left(\frac{e^{z_i/T}}{\sum_j e^{z_j/T}} - \frac{e^{v_i/T}}{\sum_j e^{v_j/T}} \right)$$

If the temperature is high, the following equation takes place:

$$\frac{\partial C}{\partial z_i} \approx \frac{1}{T} \left(\frac{1 + z_i/T}{N + \sum_j z_j/T} - \frac{1 + v_i/T}{N + \sum_j v_j/T} \right)$$

Knowledge distillation

Logits can be centered, so

$$\sum_j z_j = \sum_j v_j = 0$$

Then the gradient takes form:

$$\frac{\partial C}{\partial z_i} \approx \frac{1}{T} \left(\frac{1 + z_i/T}{N + \sum_j z_j/T} - \frac{1 + v_i/T}{N + \sum_j v_j/T} \right) \approx \frac{1}{NT^2} (z_i - v_i)$$

$$\frac{dC}{dz_i} = \frac{1}{NT^2} (z_i - v_i) \sim (z_i - v_i) \overset{\text{Constant}}{=} M \frac{d(z_i - v_i)^2}{dz_i}$$

Bias-Variance decomposition

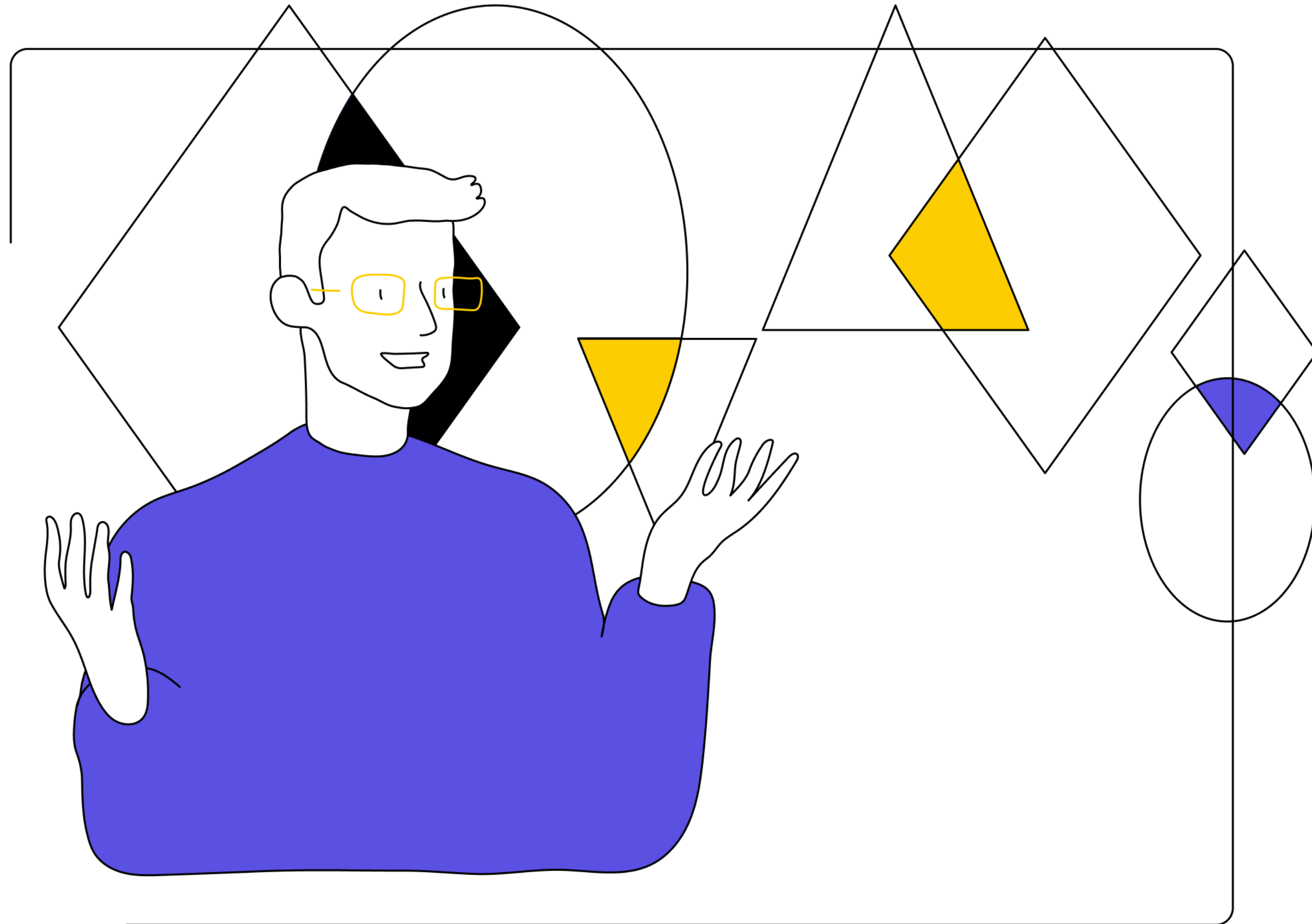
$$L(\mu) = \underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{Noise}} + \underbrace{\mathbb{E}_x \left[(\mathbb{E}_X [\mu(X)] - \mathbb{E}[y | x])^2 \right]}_{\text{Bias}} + \underbrace{\mathbb{E}_x \left[\mathbb{E}_X \left[(\mu(X) - \mathbb{E}_X [\mu(X)])^2 \right] \right]}_{\text{Variance}}$$

This exact form of bias-variance decomposition is correct for square loss in regression

However, it is much more general. See extra materials for more exotic cases

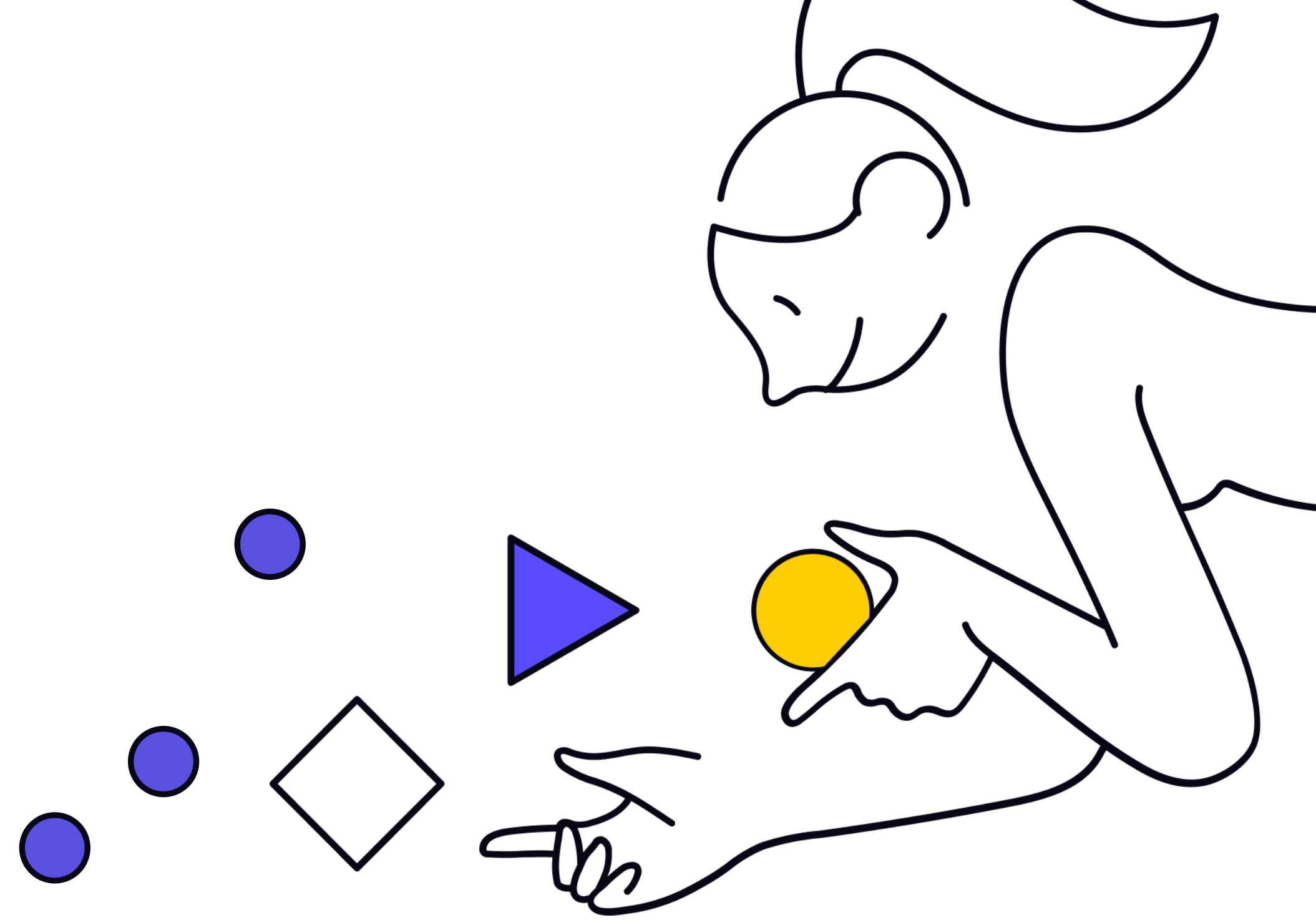
Feature importance estimation: naïve approach

02



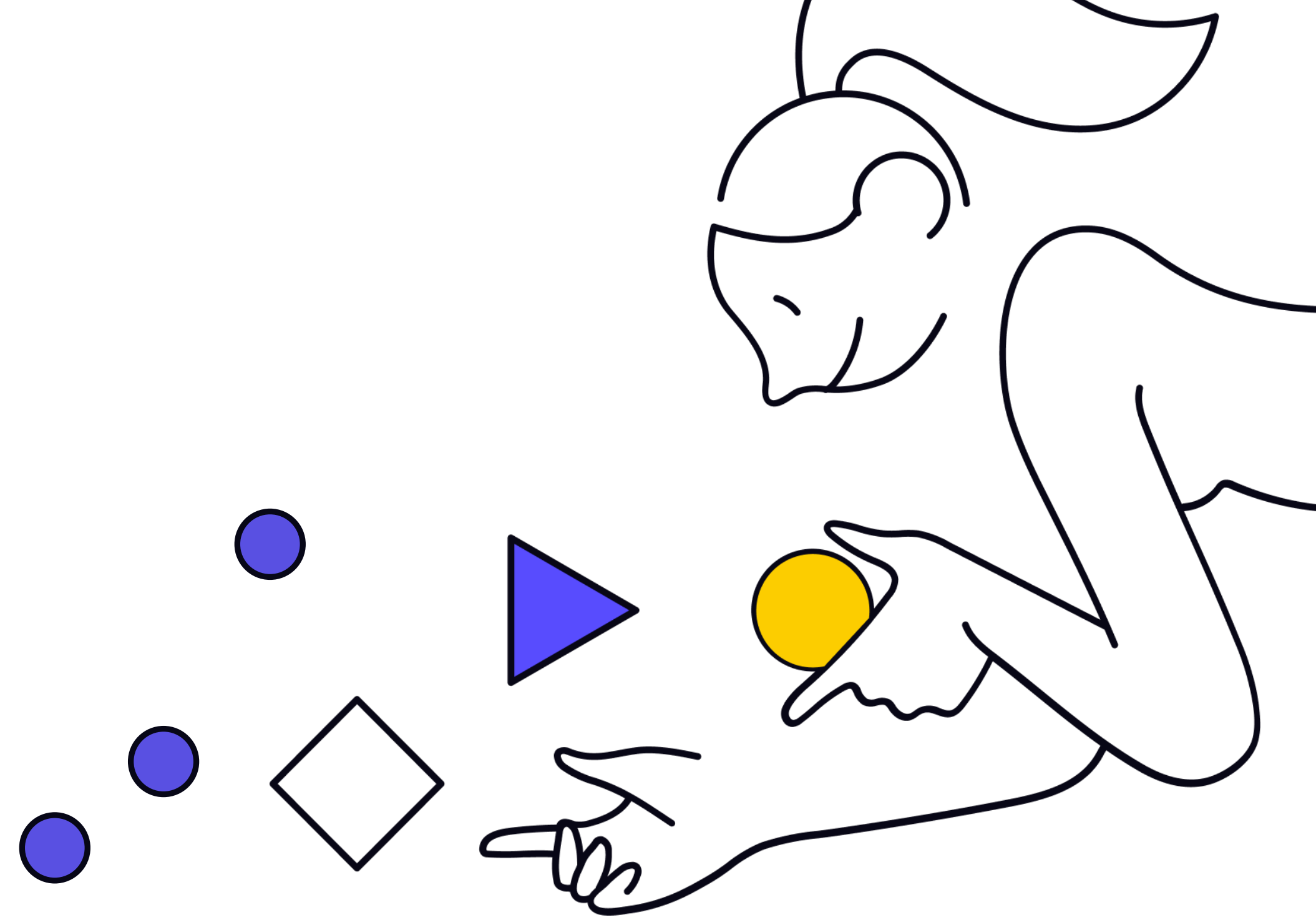
List of methods so far

1. kNN – non-parametric method



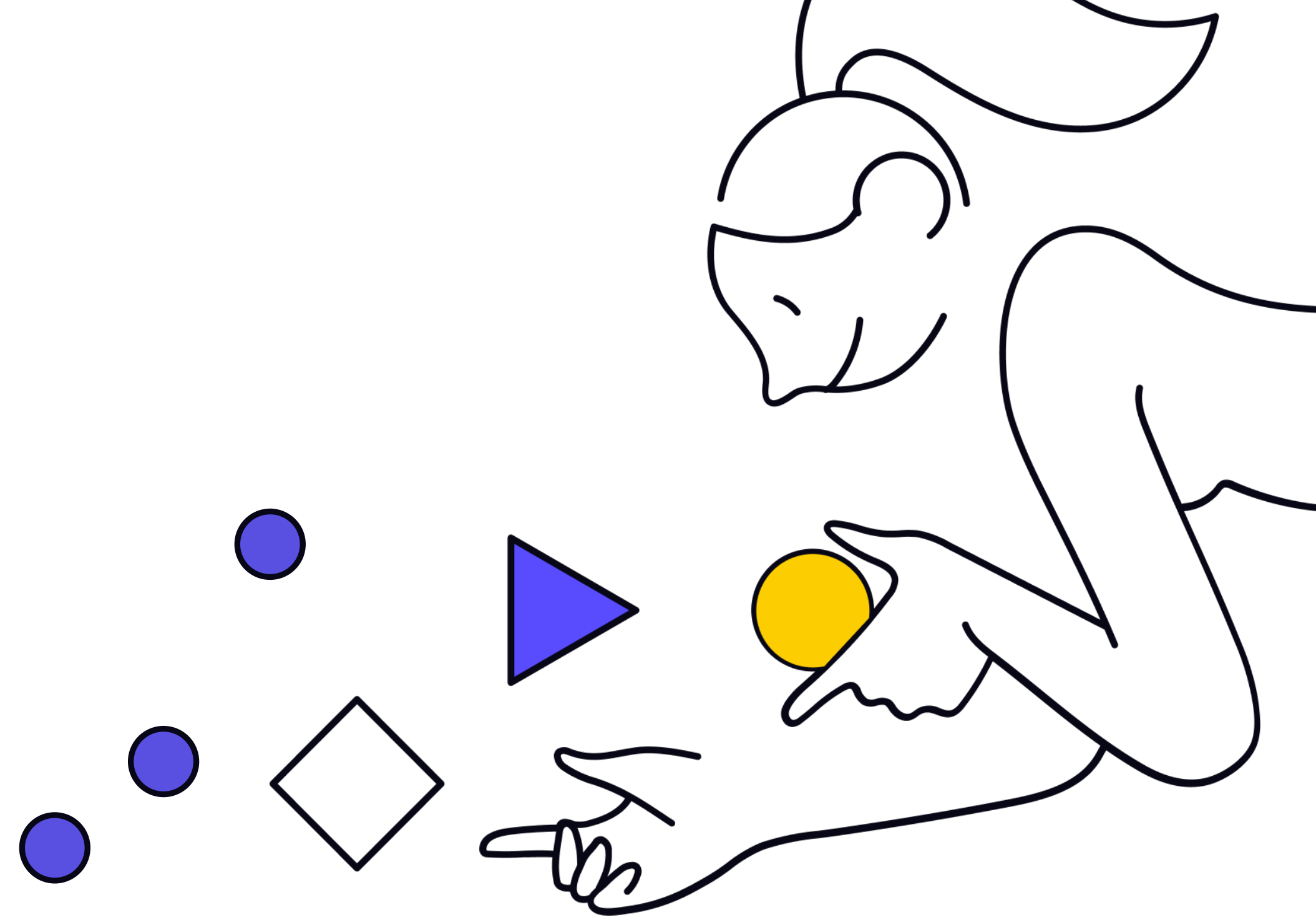
List of methods so far

1. kNN – non-parametric method
2. Linear (logistic) regression



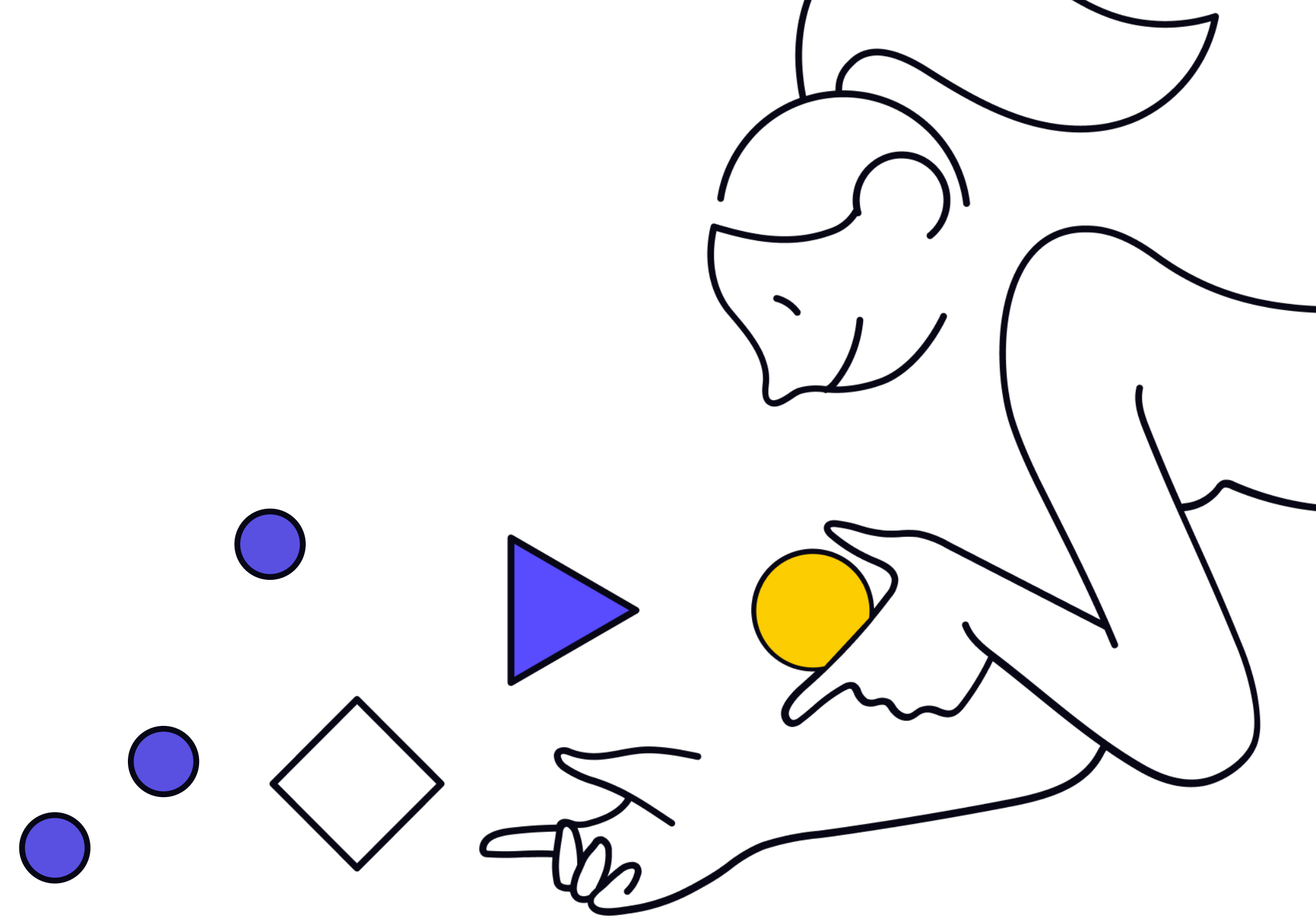
List of methods so far

1. kNN – non-parametric method
2. Linear (logistic) regression
3. Decision Trees



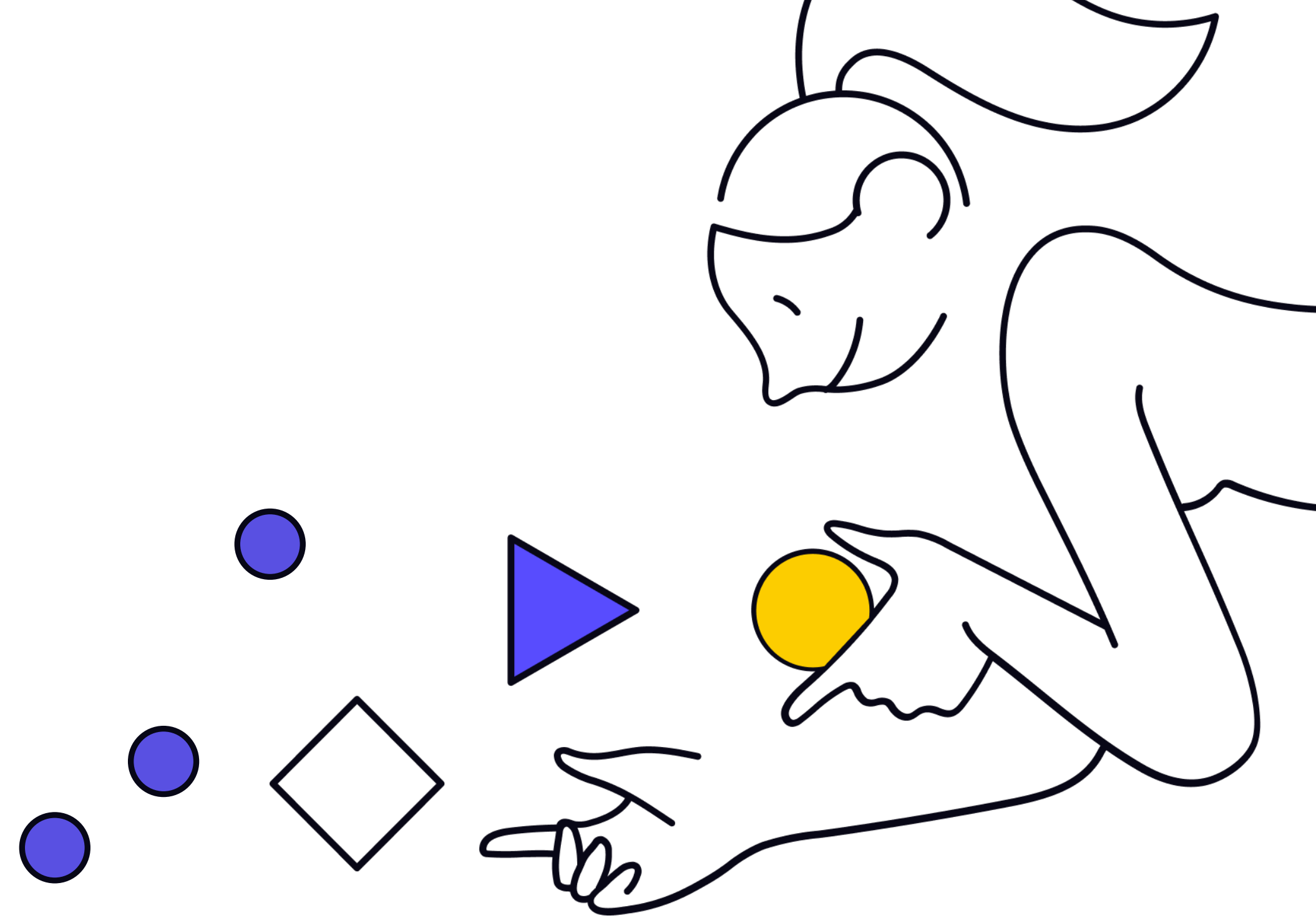
List of methods so far

1. kNN – non-parametric method
2. Linear (logistic) regression
3. Decision Trees
4. Bagging ensembles



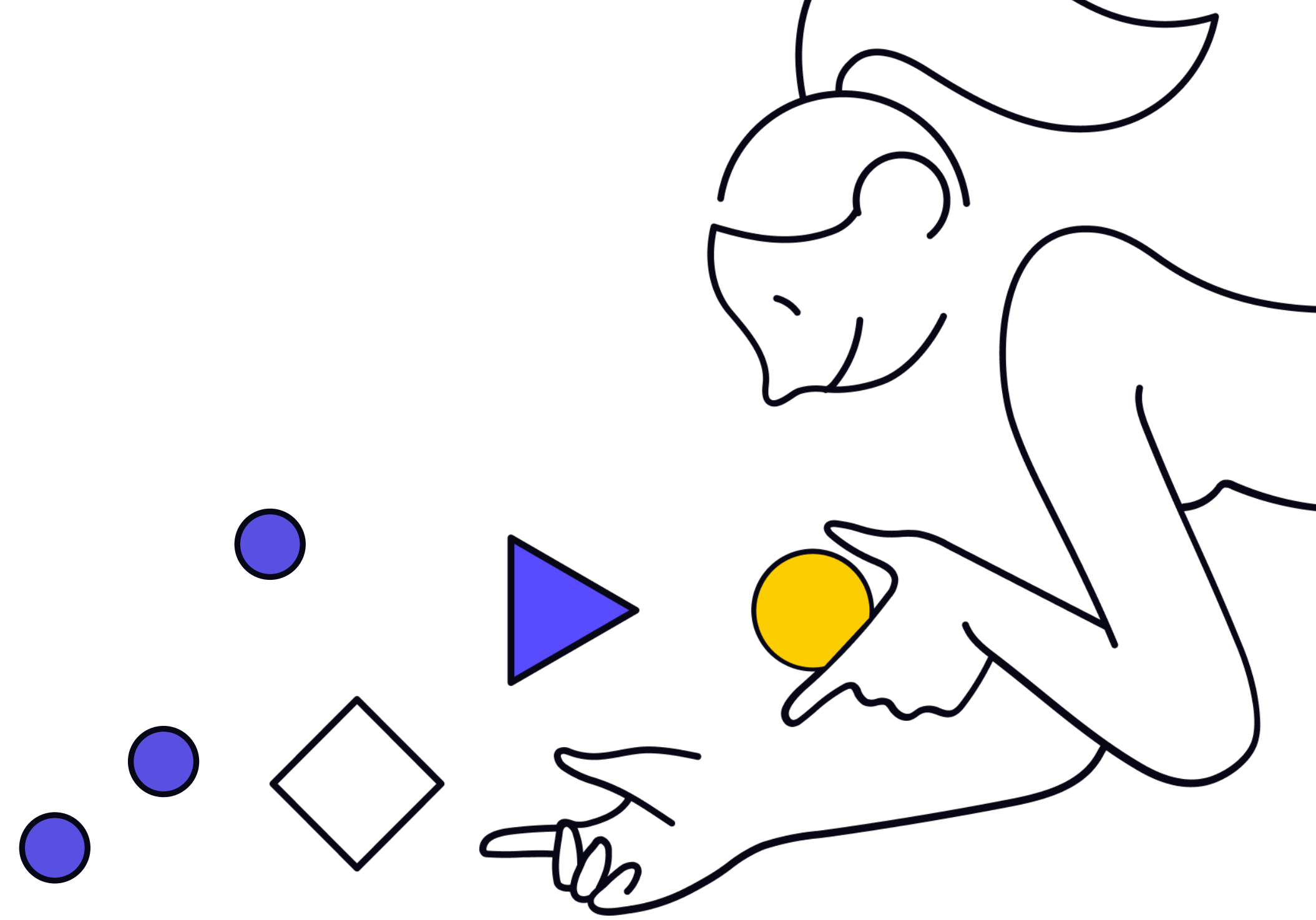
List of methods so far

1. kNN – non-parametric method
2. Linear (logistic) regression
3. Decision Trees
4. Bagging ensembles
5. Boosting ensembles



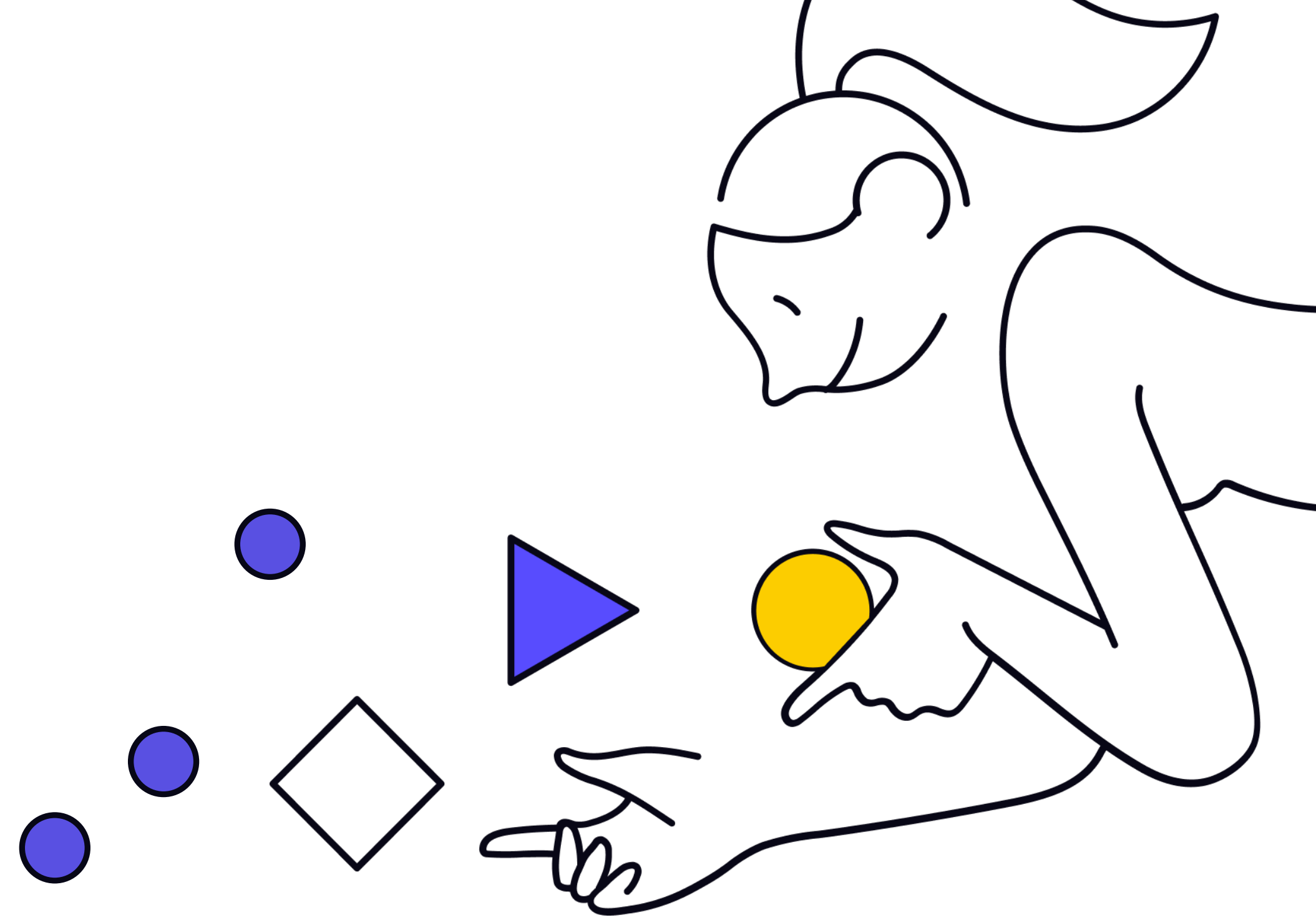
List of methods so far

1. kNN – non-parametric method
2. Linear (logistic) regression
3. Decision Trees
4. Bagging ensembles
5. Boosting ensembles
6. Stacking and blending



List of methods so far

1. kNN – non-parametric method
2. Linear (logistic) regression
3. Decision Trees
4. Bagging ensembles
5. Boosting ensembles
6. Stacking and blending
7. Neural Networks



Today's lecture recap

- 01 Bias-Variance decomposition
- 02 Feature importance estimation: naïve approach
- 03 LIME
- 04 Shapley values; SHAP
- 05 Neural networks predictions explanation



Thanks for attention!

Questions?